



INTERNATIONAL UNIVERSITY OF RABAT

Integrated Project

Retail AI Multi-Agent System

Intelligent Retail Forecasting, Anomaly Detection,
Recommendations, and Automated Business Reporting

AI-powered retail analytics platform based on collaborative agents,
Deep Learning models, and Human-in-the-Loop validation.

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Contents

1	Introduction	3
2	Project Context and Objectives	4
2.1	Project Context	4
2.2	Project Objectives	4
3	System Architecture	5
4	Implemented Agents	6
4.1	Generic Preprocessing Agent	6
4.2	LSTM Forecasting and Trend Analysis Agent	6
4.3	LSTM Autoencoder Anomaly Detection Agent	6
4.4	Recommendation Agent	6
4.5	Human-in-the-Loop Agent	7
4.6	Report Generation Agent	7
4.7	Orchestrator Agent	7
5	Deep Learning Models	7
5.1	LSTM Forecasting Model	7
5.2	Forecasting Training Process	8
5.3	LSTM Autoencoder for Anomaly Detection	8
5.4	Deep Learning Integration inside the Multi-Agent System	8
6	Workflow and User Interface	9
7	Results and Evaluation	10
7.1	Dataset Analysis and Execution	10
7.2	Forecasting Metrics	10
7.3	Future Demand Forecasting	11
7.4	Training and Validation Performance	11
8	Anomaly Detection Results	12
8.1	Anomaly Detection Pipeline	12
8.2	Anomaly Detection Metrics	12
8.3	LSTM Autoencoder Training Performance	13
9	Conclusion	13

List of Figures

1	Retail AI Multi-Agent System Architecture	5
2	Retail AI Multi-Agent System User Interface	9
3	Dataset Upload and Pipeline Execution	10
4	Forecasting Metrics and Model Evaluation	11
5	Future Demand Forecasting and Peak Detection	11
6	LSTM Training and Validation Loss	12
7	Anomaly Detection Pipeline Execution	12
8	Anomaly Detection Metrics and Risk Distribution	13
9	LSTM Autoencoder Training and Validation Loss	13

1 Introduction

Artificial intelligence is increasingly used to support business decision-making, especially in domains where large volumes of data must be analyzed quickly and accurately. In the retail sector, companies need to understand sales evolution, anticipate future demand, detect abnormal behaviors, and prepare operational actions such as inventory adjustment or promotion planning.

This project focuses on the design and implementation of a Retail AI Multi-Agent System. The objective is to build an intelligent platform where several specialized agents collaborate to analyze retail datasets. Each agent has a specific role, such as preprocessing data, forecasting sales, detecting anomalies, generating recommendations, validating decisions, and producing final reports.

The system integrates deep learning models, mainly based on LSTM architectures, to perform time-series forecasting and anomaly detection. A Human-in-the-Loop mechanism is also included to allow human validation before accepting important business recommendations. This makes the system more reliable because final decisions are not fully automated.

The project follows the official requirements of the Integrated Project, which asks for a multi-agent system with specialized agents, at least one trained deep learning model, orchestration, error handling, logging, and human validation. The final solution is applied to a retail sales dataset containing information such as dates, products, categories, prices, promotions, units sold, and inventory levels.

2 Project Context and Objectives

2.1 Project Context

Retail companies generate large amounts of transactional and operational data every day. This data contains valuable information about customer demand, sales evolution, promotions, and inventory management. However, manually analyzing these datasets becomes difficult when the data grows in volume and complexity.

Traditional business analysis methods are often limited because they cannot easily predict future demand or automatically detect abnormal situations such as unexpected sales spikes, stock shortages, or inconsistent promotional behaviors. For this reason, artificial intelligence and deep learning techniques are becoming important tools in modern retail analytics.

In this project, we developed an intelligent multi-agent platform capable of automating several retail analysis tasks. The system uses collaborative AI agents where each agent performs a dedicated responsibility inside the global workflow. The platform combines forecasting, anomaly detection, recommendation generation, and automated reporting in a unified architecture.

The project was implemented following the integrated project requirements of the AI & Big Data program at UIR, including multi-agent collaboration, deep learning integration, orchestration, logging, and Human-in-the-Loop validation.

2.2 Project Objectives

The main objective of this project is to design and implement a complete Retail AI Multi-Agent System capable of assisting retail decision-making through intelligent data analysis.

The system aims to:

- Prepare and preprocess retail datasets automatically
- Forecast future retail demand using LSTM deep learning models
- Detect abnormal sales and inventory behaviors
- Generate intelligent business recommendations
- Integrate human validation before final decisions
- Produce automated business intelligence reports
- Coordinate all analysis steps using an orchestrator agent

The final objective is to create an intelligent and modular retail analytics platform that improves forecasting accuracy, anomaly detection, and operational decision-making.

3 System Architecture

The proposed solution is based on a collaborative multi-agent architecture where several intelligent agents work together to accomplish retail analytics tasks through a centralized orchestration workflow.

The process starts when the user uploads retail CSV datasets through the graphical interface. The Orchestrator Agent manages communication and execution between all agents.

The Preprocessing Agent cleans and prepares the dataset for deep learning models. Depending on the selected analysis mode, the system launches either the LSTM Forecasting Agent for future demand prediction and peak detection, or the Anomaly Detection Agent for abnormal retail behavior analysis.

The generated outputs are then analyzed by the Recommendation Agent to produce operational recommendations and business insights. Before generating the final report, the Human-in-the-Loop Agent validates important recommendations manually.

Finally, the Report Generation Agent creates a structured business intelligence report containing forecasting results, anomaly summaries, recommendations, and visualizations.

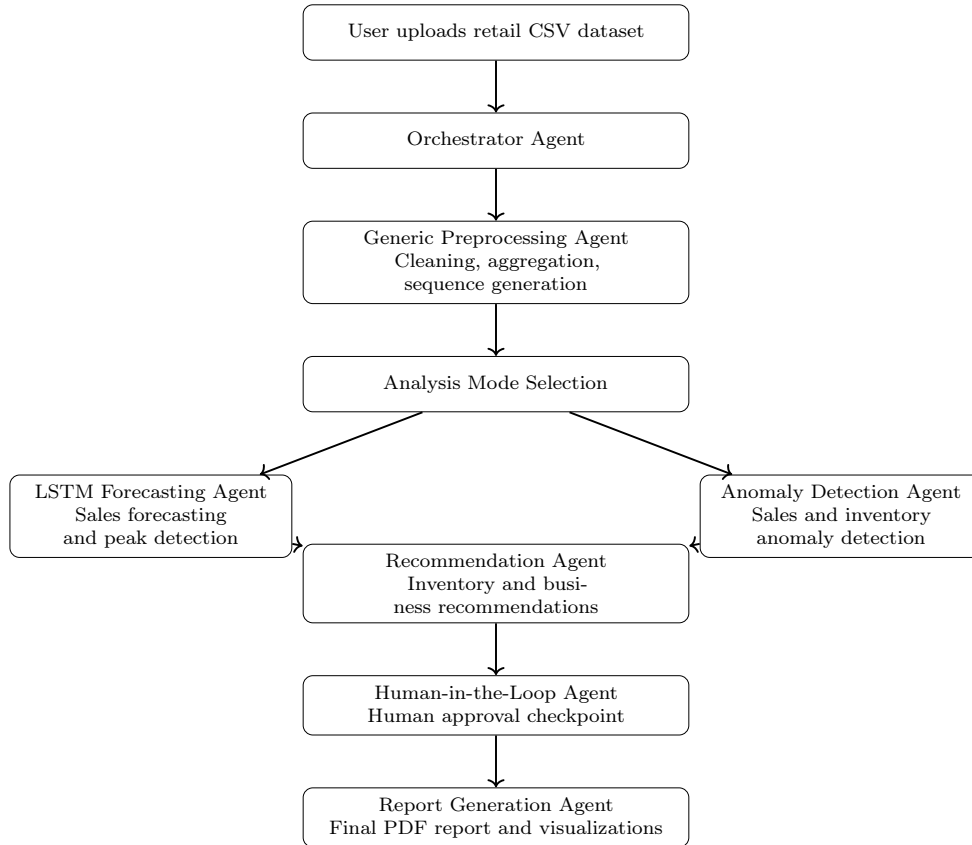


Figure 1: Retail AI Multi-Agent System Architecture

4 Implemented Agents

The proposed system is composed of several specialized agents collaborating through a coordinated workflow. Each agent has a dedicated responsibility and contributes to the global retail analysis pipeline.

4.1 Generic Preprocessing Agent

The Generic Preprocessing Agent prepares the uploaded retail dataset before model execution. It loads CSV files, detects the date column and target sales column, removes invalid rows and duplicates, handles missing values, creates time-based features, and produces a model-ready daily time-series dataset.

The main outputs of this agent are a clean merged dataset and a model-ready dataset used by the forecasting and anomaly detection models.

4.2 LSTM Forecasting and Trend Analysis Agent

The Forecasting Agent uses a PyTorch LSTM model to predict future retail demand. The model receives time-series sequences built from historical sales and engineered features such as day, month, weekend indicator, lag values, moving averages, exponential moving averages, and percentage change.

After training, the agent evaluates the model using metrics such as MAE, RMSE, R2 score, and SMAPE. It also predicts future sales, detects peak demand periods, analyzes demand trends, and estimates recommended inventory using a safety margin.

4.3 LSTM Autoencoder Anomaly Detection Agent

The Anomaly Detection Agent uses an LSTM Autoencoder to learn normal temporal retail behavior. The model reconstructs input sequences, and abnormal periods are identified using reconstruction error.

This agent also combines deep learning results with business rules such as sales spikes, sales drops, z-score thresholds, and suspicious promotion behavior. Each anomaly is assigned a type and a risk level such as low, medium, or high risk.

4.4 Recommendation Agent

The Recommendation Agent transforms the forecasting and anomaly outputs into business recommendations. In forecasting mode, it recommends actions such as increasing stock, preparing inventory before peak demand dates, or adapting promotion strategies. In anomaly mode, it recommends investigating abnormal sales periods and validating suspicious situations before operational decisions.

This agent uses rule-based logic and can also call the CrewAI/Ollama layer to generate a business insight summary.

4.5 Human-in-the-Loop Agent

The Human-in-the-Loop Agent validates the generated recommendations before final reporting. It reviews the priority of each recommendation, approves it, and adds reviewer comments depending on the risk level. This step ensures that important business actions are not accepted without human validation.

4.6 Report Generation Agent

The Report Generation Agent creates the final business report. It summarizes forecasting or anomaly detection results, business recommendations, CrewAI insights, human validation results, and final business conclusions. The report is saved in structured JSON format and also as a readable text report.

4.7 Orchestrator Agent

The Orchestrator Agent coordinates the complete pipeline. It starts preprocessing, checks the required saved model files, runs either forecasting inference or anomaly detection inference, launches the recommendation agent, sends outputs to the Human-in-the-Loop Agent, and finally starts report generation.

It also logs each step with timestamps and handles errors to prevent the pipeline from crashing.

5 Deep Learning Models

5.1 LSTM Forecasting Model

The forecasting component is based on a Long Short-Term Memory (LSTM) neural network implemented with PyTorch. LSTM models are well adapted for time-series forecasting because they can learn temporal dependencies from historical retail data.

The model receives sequences generated during preprocessing and uses engineered features such as lag values, moving averages, percentage changes, and calendar-based information.

The forecasting architecture contains:

- One LSTM layer
- Fully connected dense layers
- ReLU activation
- Dropout regularization

The model predicts future sales demand and identifies future peak periods.

5.2 Forecasting Training Process

The forecasting model was trained using retail time-series data divided into training, validation, and test sets.

The training process uses:

- AdamW optimizer
- Huber Loss function
- Early stopping
- Gradient clipping
- Learning rate scheduling

The forecasting performance was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R2 Score
- SMAPE

5.3 LSTM Autoencoder for Anomaly Detection

The anomaly detection component uses an LSTM Autoencoder implemented with PyTorch. The model learns normal retail behavior by reconstructing input sequences. High reconstruction error indicates abnormal retail periods.

The anomaly detection process combines:

- Reconstruction error analysis
- Z-score analysis
- Sales spike and drop detection
- Promotion anomaly analysis

Each anomaly is assigned a type, risk level, and business interpretation.

5.4 Deep Learning Integration inside the Multi-Agent System

The deep learning models are fully integrated into the multi-agent architecture. Depending on the selected analysis mode, the Orchestrator Agent launches either the forecasting model or the anomaly detection model.

The generated outputs are then transmitted to the Recommendation Agent and the Report Generation Agent for business analysis and reporting.

6 Workflow and User Interface

The system execution begins through a Streamlit graphical user interface designed to simplify interaction with the multi-agent platform.

The interface allows the user to:

- Upload one or multiple retail CSV datasets
- Select the analysis mode
- Execute forecasting or anomaly detection inference
- Visualize generated results
- Download final business reports

After the dataset is uploaded, the Orchestrator Agent launches the preprocessing pipeline automatically. The preprocessing phase cleans the dataset, creates temporal features, aggregates retail sales information, and generates model-ready sequences.

The user can then select one of the following analysis modes:

- Forecasting and Peak Detection
- Anomaly Detection

In forecasting mode, the LSTM forecasting model predicts future retail demand and identifies future peak periods. In anomaly detection mode, the LSTM Autoencoder analyzes temporal retail behavior and detects abnormal patterns.

The generated outputs are transmitted to the Recommendation Agent, which creates business recommendations and operational insights. These recommendations are then validated by the Human-in-the-Loop Agent before the final report is generated.

The complete workflow is monitored by the Orchestrator Agent, which manages execution order, communication between agents, logging, and error handling.

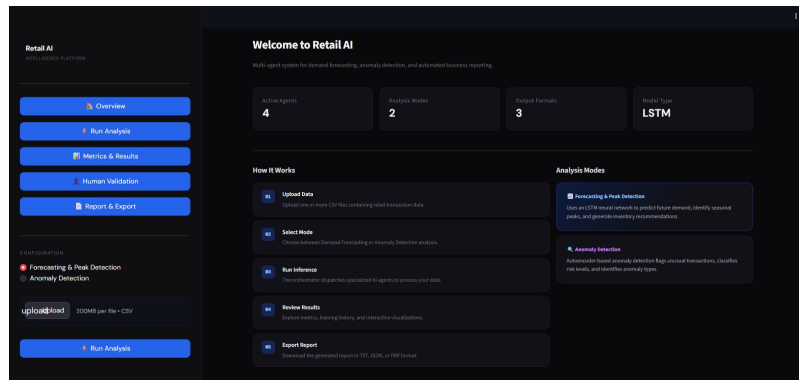


Figure 2: Retail AI Multi-Agent System User Interface

7 Results and Evaluation

The developed system was successfully tested on retail sales datasets containing transactional and inventory information. The experiments demonstrate the ability of the multi-agent system to preprocess retail data, forecast future demand, detect anomalies, generate recommendations, and produce automated business reports.

7.1 Dataset Analysis and Execution

The uploaded retail dataset contains information related to products, stores, promotions, inventory levels, and sales quantities. After preprocessing, the data was transformed into structured time-series sequences compatible with the deep learning models.

The execution pipeline successfully completed the preprocessing phase, model inference, recommendation generation, human validation, and final report generation.

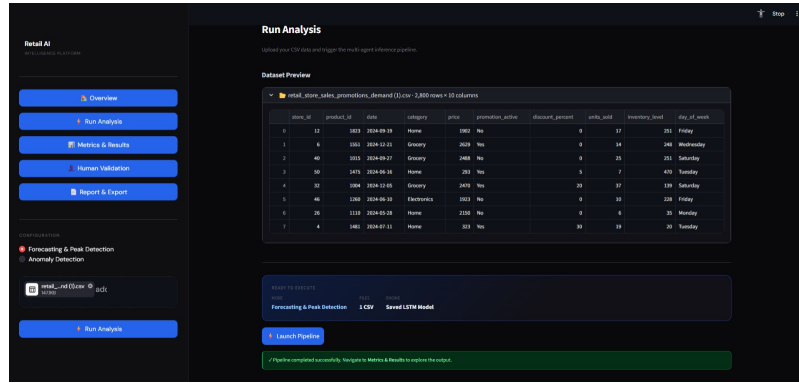


Figure 3: Dataset Upload and Pipeline Execution

7.2 Forecasting Metrics

The forecasting model successfully generated future retail demand predictions and evaluation metrics. The obtained results show that the LSTM model was capable of learning temporal retail patterns and generating stable demand forecasts.

The evaluation dashboard summarizes several important forecasting indicators, including MAE, RMSE, R2 Score, and SMAPE. The system also estimates future demand variation, total predicted demand, recommended inventory levels, and detected peak periods.

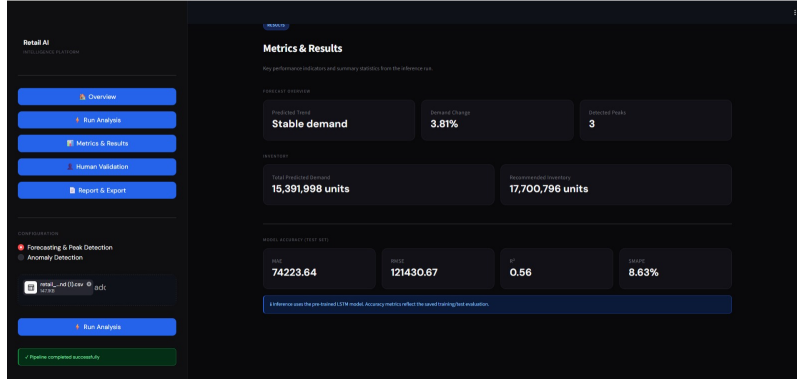


Figure 4: Forecasting Metrics and Model Evaluation

7.3 Future Demand Forecasting

The forecasting model predicted future sales demand and automatically identified peak demand periods. The detected peaks can help retail managers prepare inventory and operational resources in advance.

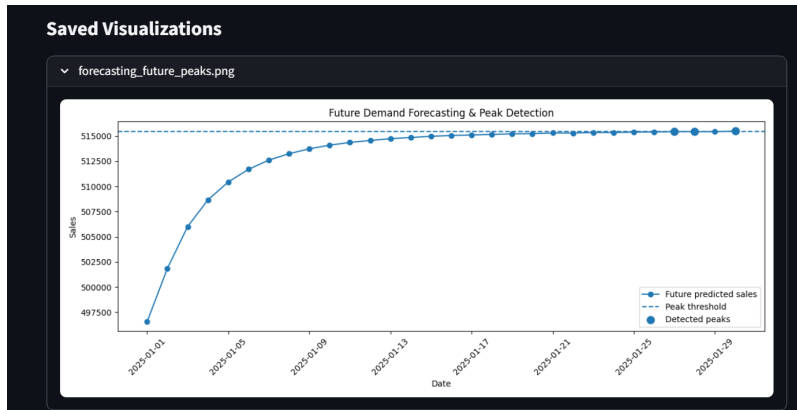


Figure 5: Future Demand Forecasting and Peak Detection

7.4 Training and Validation Performance

The following figure presents the evolution of training and validation loss during the LSTM training process. The gradual decrease of both losses indicates that the model successfully learned temporal retail patterns while maintaining acceptable generalization performance.

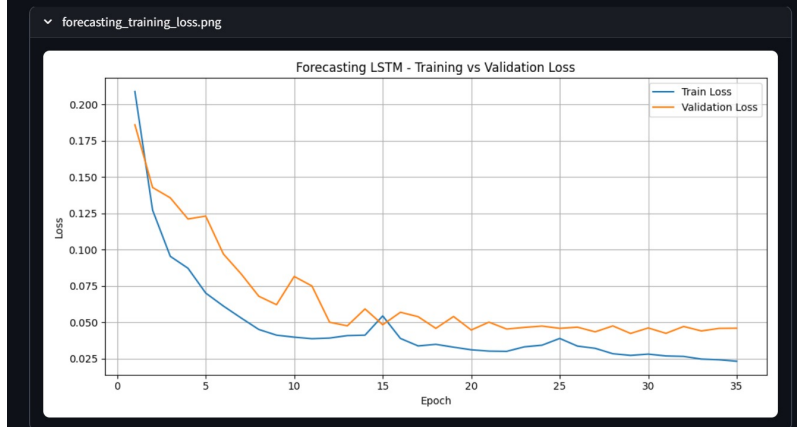


Figure 6: LSTM Training and Validation Loss

8 Anomaly Detection Results

The anomaly detection pipeline was successfully executed using the LSTM Autoencoder model. The system analyzed historical retail sales behavior and identified abnormal temporal patterns based on reconstruction error and business anomaly rules.

8.1 Anomaly Detection Pipeline

The anomaly detection workflow starts by preprocessing retail sales sequences and loading the trained LSTM Autoencoder model. The model reconstructs normal sales behavior and computes reconstruction error values for anomaly identification.

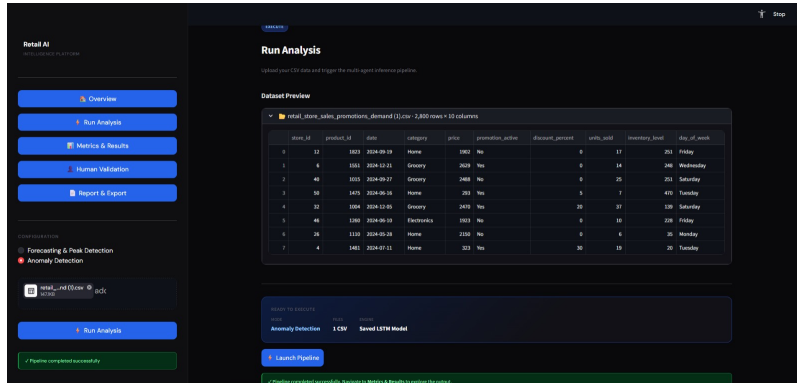


Figure 7: Anomaly Detection Pipeline Execution

8.2 Anomaly Detection Metrics

The anomaly detection system generated several performance indicators related to reconstruction loss, anomaly threshold estimation, anomaly rate, and detected anomaly categories.

The obtained results demonstrate that the model was capable of identifying inconsistent demand behaviors and unusual retail patterns.

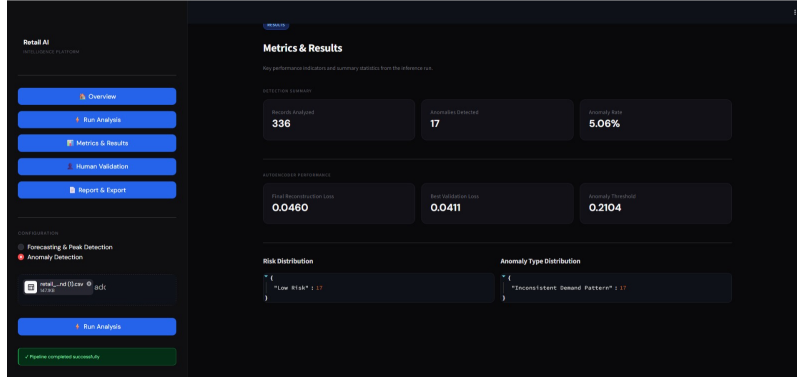


Figure 8: Anomaly Detection Metrics and Risk Distribution

8.3 LSTM Autoencoder Training Performance

The following figure presents the evolution of training and validation loss during the training process of the LSTM Autoencoder model.

The convergence of both curves indicates that the model successfully learned normal temporal retail behavior while maintaining stable validation performance.

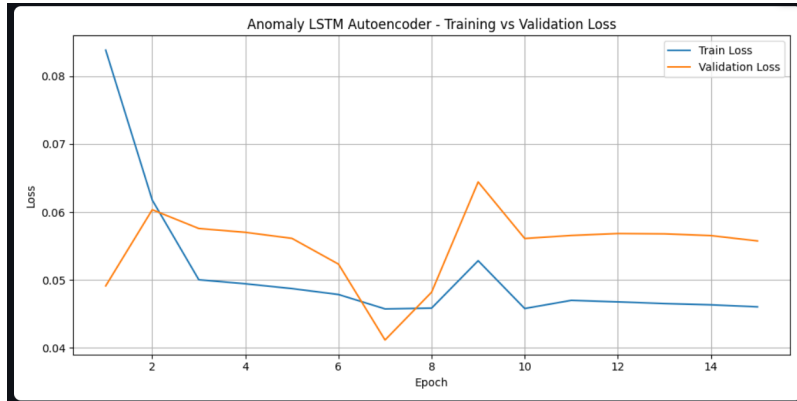


Figure 9: LSTM Autoencoder Training and Validation Loss

9 Conclusion

This project presented the implementation of a Retail AI Multi-Agent System for intelligent retail analytics and business decision support.

The developed platform integrates multiple specialized agents responsible for preprocessing, forecasting, anomaly detection, recommendation generation, human validation, and automated reporting. LSTM-based deep learning models were successfully used to analyze temporal retail data and predict future demand behavior.

The obtained results demonstrate the ability of the system to identify sales trends, detect abnormal retail behaviors, generate operational recommendations, and automate business reporting through a collaborative multi-agent workflow.